Reg. No. | | | | | | | | | |

Question Paper Code: 14108

B.E. / B.TECH DEGREE EXAMINATION, APRIL / MAY 2018

Third Semester

B.E. – Computer Science and Engineering

15CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2015)

Duration: 3 Hours Max.: 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Write the purpose of MAR and MDR.
2. Let X=1010100 and Y=1000011 perform
   A) X – Y and
   B) Y – X using 2's complement.
3. Write the register transfer sequence to read a word from memory.
4. What is hardwired control? How is it different from micro-programmed control?
5. Define pipeline speedup and efficiency.
6. What is meant by delayed branching?
7. An address space is specified by 24 bits and the corresponding memory space by 16bits:
   How many words are there in the virtual memory and in the main memory?
8. Define the terms hit and miss rate with respect to cache.
9. Distinguish between isolated and memory mapped I/O.
10. Compare synchronous and asynchronous buses.

PART B – (5 × 16 = 80 Marks)

11. a) i. Using carry-save addition of summands multiply
   A) the multiplicand $M=101011_2$ by the multiplier $Q=101101_2$
   B) the multiplicand $M=110101_2$ by the multiplier $Q=100011_2$ (8 Marks)

11. a) ii. Explain different types of instructions with examples. Compare their relative merits and demerits. (8 Marks)

OR

11. b) i. Assuming 6 – bit 2's complement number representation, multiply the multiplicand A=110011 by the multiplier B=010101 using Booth algorithm and bit pair recoding of multipliers. (8 Marks)

11. b) ii. Why do we use addressing mode? Explain the basic addressing modes. (8 Marks)

12. a) i. What are the sequence of actions needed to fetch and execute the instruction:

A) Load R5, X (R7)

B) Store R6, X (R8) (8 Marks)

12. a) ii. Short notes on

A) ALU

B) Datapath (8 Marks)

OR

12. b) i. Give the sequences of actions for a Return – from – subroutines instructions in a RISC processor. Assume that the address LINK of the general – purpose register in which the subroutine return address is stored is given in the instruction filed connected to address A of the register file (IR_{31-27}). Justify the same with PC-update hardware support. (8 Marks)

12. b) ii. Draw the necessary diagrams and explain the control signal generation using hardwired control. (8 Marks)

13. a) Explain the different types of hazards that can occur in a pipeline. (16 Marks)

OR

13. b) Discuss in detail about the pipelining in CISC processor. (16 Marks)

14. a) i. Discuss Direct Memory Access in detail. (10 Marks)

14. a) ii. A computer system uses 32-bit memory addresses and it has a main memory consisting of 1 Gbytes. It has a 4K-byte cache organized in the block-set-associative manner, with 4 blocks per set and 64 bytes per block.

Calculate the number of bits in each of the Tag, Set, and Word fields of the memory address. (6 Marks)

OR

14. b) i. Explain the virtual memory address translation with necessary tables and diagrams. (10 Marks)

14. b) ii. Design a 128Kx8 memory chip with required external connections. (6 Marks)

15. a) Discuss the Interconnection standards:

A) HDMI

B) PCI Bus (16 Marks)

OR

15. b) i. What are the types of interrupt? Explain how the interrupts are handle based on its priority? (8 Marks)

15. b) ii. What is program controlled I/O? Explain how input and output operations are controlled with registers support. (8 Marks)